Practical - 6

Rendering External JSON Data using Fetch API, Search and Filter

A . Theory

Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.

1 How is JSON fetched, parsed, and rendered?

We use `fetch()` to get the JSON file. Then `.json()` parses the response into a JavaScript object/array. After that, we use JavaScript and DOM methods like `innerHTML` to display the data on the webpage.

Which array methods are used for search, filter, sort, and pagination?

- **Search:** `filter()` — finds matching records.
- **Filter:** `filter()` — selects records based on a condition.
- **Sort:** `sort()` — arranges records in ascending/descending order.
- **Pagination:** `slice()` — displays only a specific portion of the array.

2 How are loading and error states handled?

Initially, we display a **"Loading..."** message. If the JSON is successfully fetched, the data replaces the loading message. If an error occurs, `.catch()` displays an **error message** to the user.

3 How is modularity maintained in JavaScript files?

We divide functionality into separate JavaScript files or functions. For example, one function can handle fetching data, another rendering, and others can handle search, sorting, filtering, and pagination. This makes the code easier to understand, maintain, and reuse.

B. Implementation/ Output/Result

Here is my GitHub repo link the codes are here in repo:
https://github.com/dhyaani19/StudentHub-Portal.git

C.. Evidence of Work (Screenshots)

Figure 6.1 : Home page (index.html)

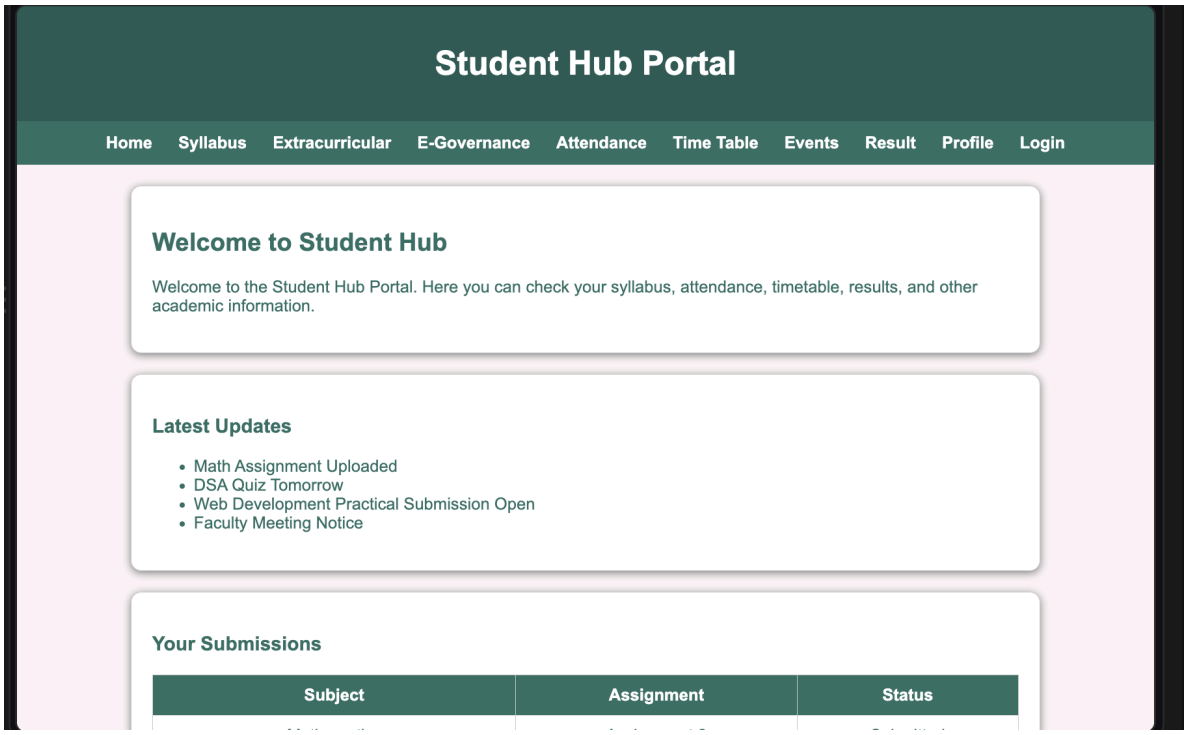


Figure 6.2 : Home page (index.html)

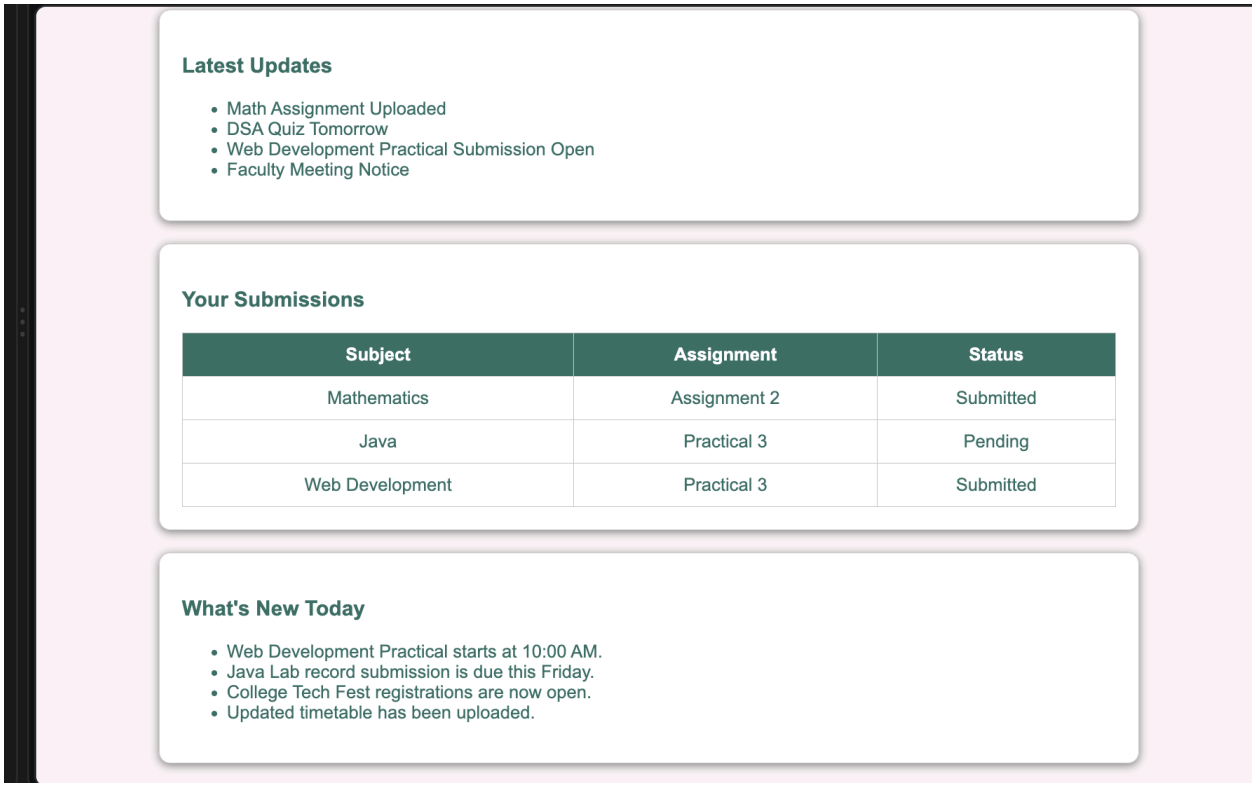


Figure 6.3: Event page (event.html)

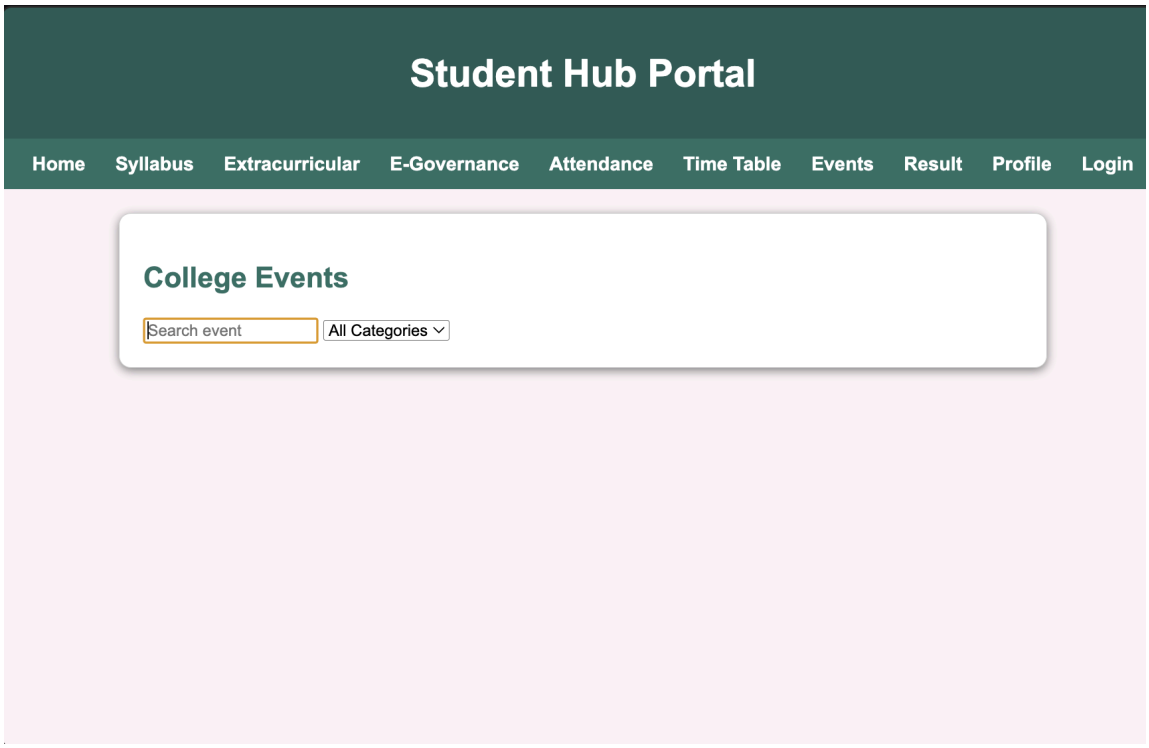
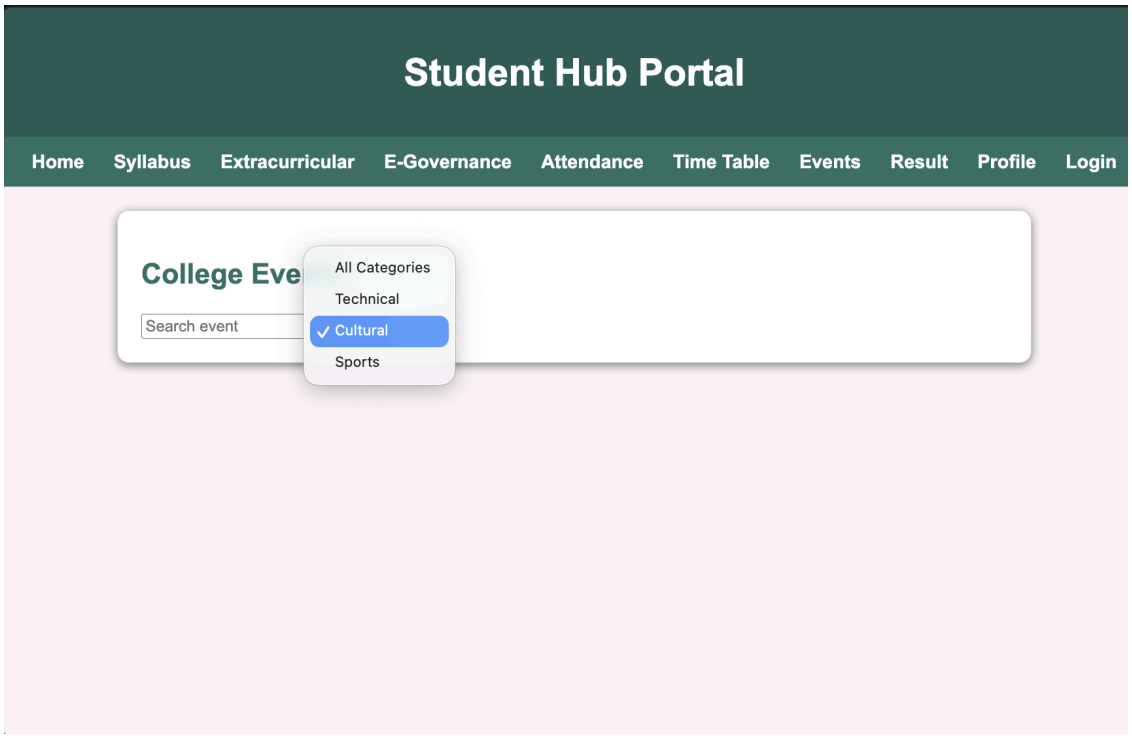


Figure 6.4:Event page (event.html)



D. GitHub Repository Link:

https://github.com/dhyaani19/StudentHub-Portal.git

E. Challenges Faced:

While doing this practical, I faced some difficulties in fetching the JSON file and giving the correct file path. At first, I was confused about how to use fetch() and convert the JSON data into JavaScript data. Implementing search, filter, sort and pagination was also a little difficult. I also faced some problems while displaying the data dynamically on the webpage and handling errors. After practicing and correcting the code, I was able to understand how all these features work together.

F. Learning Outcomes:

In this practical, I learned how to fetch data from a JSON file using JavaScript and display it on the webpage. I also learned how to search, filter and sort the data using different array methods. I understood how pagination works and how to show data page by page. I also learned how to handle loading and error messages when data is not loaded properly. Overall, this practical helped me improve my understanding of JavaScript, JSON and DOM manipulation.

Faculty Remarks: